

Evidence Model, Provenance and Decision States

Scientific objects

Species + assembly
source record identifiers

Gene -> transcript -> exon / CDS
strand + CDS phase

Translated protein
transcript-protein identity

Inspectable evidence record

Biological identity
species - transcript - protein - exon / junction

Boundary observation
domain_instance_id - interval - source
nearest edge - distance - class - threshold policy

Context and QC
comparable group - analysis scope
projection / annotation status - review reason

Coordinate evidence register
exon-protein | domain | transmembrane | boundary relation
protein_aa_1_based_inclusive

Evidence-producing activities

Genome-to-protein projection
ordered CDS + strand + phase

Annotation normalization
ownership + coordinate conversion + layer

Nearest-edge measurement
signed + absolute amino-acid distance

Provenance and decisions

Source-row and run provenance
source table + row - model link - run reports
source-file SHA-256 + generation metadata

Run lifecycle
created - processing - results ready / partial - failed

Analysis availability
available - not applicable - pending - scientifically unavailable
technically missing - stale - failed

Scientific evidence state
exploratory or validated | primary result or review / missing

The additive register is generated before and after cluster annotation: domain-dependent evidence is pending_cluster before annotation and refreshed afterward.
Scientific values are copied from their source tables; unavailable metadata is marked explicitly and is not inferred silently.